The No-Signaling Theorem under a Minimalist Event-Based Theory of Information

Sagar Kapadia[1*]

[1*]Independent Researcher, C-33/35, Vaibhav Apartment, Athwa Lines, Surat, 395001, Gujarat, India.

Corresponding author(s). E-mail(s): sagar.kapadia@hotmail.com;

Abstract

The no-signaling theorem asserts that quantum entanglement cannot be used to transmit controllable information between spacelike-separated parties, preserving relativistic causality [1]. This paper introduces the Minimalist Event-Based Theory of Information (MEBTI), which defines information as the reliable detection of a binary event deliberately triggered or not triggered at will by an agent. Under MEBTI, we re-evaluate the no-signaling theorem's universality, arguing that its reliance on classical definitions of information (e.g., Shannon's ensemble statistics or von Neumann's state uncertainty [**?**]) renders it ontologically limited. MEBTI bypasses these constraints by focusing on single, agent-triggered events, challenging the theorem's applicability to all conceivable definitions of information. While the theorem remains a valid physical constraint within standard quantum mechanics, we propose new experiments designed under MEBTI to empirically test whether instantaneous information transmission via entanglement is possible, potentially reshaping our understanding of quantum non-locality.

Keywords: No-signaling theorem, quantum entanglement, MEBTI, information transmission, non-locality

1 Introduction

The no-signaling theorem is a cornerstone of quantum mechanics, stating that quantum entanglement cannot facilitate faster-than-light (FTL) communication between spacelike-separated parties [1**?**]. This principle ensures consistency with relativistic causality by demonstrating that local measurement choices do not affect distant marginal statistics [2]. However, the theorem's scope hinges on traditional definitions

of information, such as Shannon's reduction of ensemble uncertainty [3] or von Neumann's state entropy [**?**]. This paper introduces the Minimalist Event-Based Theory of Information (MEBTI), which redefines information as the reliable detection of a binary event triggered at will by an agent. By reframing information, MEBTI challenges the universality of the no-signaling theorem's prohibition on all forms of information transfer via entanglement. We review the literature, examine standard interpretations prohibiting FTL transmission, present MEBTI's conceptual framework, and propose experiments to test its implications.

2 Literature Review

The no-signaling theorem emerged from efforts to understand quantum entanglement, beginning with the Einstein-Podolsky-Rosen (EPR) paradox [4]. EPR argued that quantum mechanics is incomplete unless it permits non-local correlations or hidden variables, challenging the notion of "spooky action at a distance" [4]. Bell formalized this debate with inequalities that local hidden variable theories must satisfy, which quantum mechanics violates [2]. Experimental tests, such as those by Aspect et al. [5] and later loophole-free studies [6, 7], confirmed these violations, establishing quantum non-locality.

The no-signaling theorem was rigorously formulated to show that entanglement correlations do not allow controllable information transfer [1]. Peres demonstrated that marginal probabilities in entangled systems remain independent of distant measurement choices, a result derived from the linearity of quantum mechanics and the partial trace [1]. Nielsen and Chuang further contextualized this within quantum information theory, emphasizing that entanglement alone cannot serve as a communication channel [**?**]. These works rely on information defined as statistical ensemble properties, such as Shannon entropy [3] or von Neumann entropy [**?**].

Alternative theories, such as dynamical collapse models, explore modifications to quantum mechanics that might permit signaling under specific conditions [8]. However, these remain speculative and unverified. The standard no-signaling theorem remains robust within orthodox quantum mechanics, supported by empirical evidence from Bell tests [6, 7].

3 Standard Interpretations Prohibiting FTL Transmission

The no-signaling theorem asserts that the marginal statistics of one subsystem in an entangled pair are independent of the measurement choices made on the other, ensuring no FTL communication [1]. For a bipartite system with density operator ρ_{AB}, a measurement at B with operators $\{M_b^{(y)}\}$ for choice y yields the reduced state at A:

$$\rho_A^{(y)} = \sum_b \mathrm{Tr}_B \left[(I_A \otimes M_b^{(y)}) \rho_{AB} (I_A \otimes M_b^{(y)\dagger}) \right].$$

Linearity and the completeness relation $\sum_b {M_b^{(y)}}^\dagger M_b^{(y)} = I_B$ ensure $\rho_A^{(y)} = \mathrm{Tr}_B(\rho_{AB}) = \rho_A$, independent of y [**?**]. This guarantees that local measurements at A cannot detect changes in B's settings, prohibiting FTL signaling.

Bell experiments, such as those by Weihs et al. [9] and Hensen et al. [7], reinforce this by adjusting detector settings faster than light-travel time (e.g., ¡1.3 μs for a 400 m baseline [9]). The observed correlations violate Bell inequalities (e.g., CHSH: $|S| \leq 2$, quantum: $S = 2\sqrt{2}$) but maintain marginal independence, supporting the no-signaling theorem [6].

Standard interpretations rely on information as a statistical construct. Shannon's information theory defines information as the reduction of uncertainty in an ensemble, measured by entropy [3]. Von Neumann's quantum entropy similarly quantifies state uncertainty [**?**]. Gabor's perspective ties information to signal degrees of freedom [10]. These frameworks assume that information transfer involves manipulating probability distributions, which the no-signaling theorem prohibits via entanglement alone.

4 Our Contribution: A Minimalist Event-Based Theory of Information (MEBTI)

4.1 MEBTI: Definition of Information

MEBTI redefines information as the reliable detection of a binary event $E \in \{0, 1\}$, deliberately triggered or not triggered at will by an agent, within a synchronized clock interval n. Formally:

$$E_t = \begin{cases} 1 & \text{if event detected within } [t, t+n) \\ 0 & \text{otherwise} \end{cases}$$

The entropy of this event, assuming equal probabilities, is:

$$H(E) = -\sum_{i=0}^{1} p_i \log_2 p_i = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1 \text{ bit},$$

where $p_0 = p_1 = 0.5$ [3]. Unlike Shannon's ensemble-based entropy or von Neumann's state-based entropy, MEBTI focuses on a single, structureless event, prioritizing agent intentionality over statistical aggregates [3**?**].

Transmission is defined as the reliable detection of such an event across m redundant channels, using majority voting to ensure robustness. The false-positive probability is:

$$P_{FP,maj} = \sum_{k=\lceil m/2 \rceil}^{m} \binom{m}{k} p_f^k (1 - p_f)^{m-k},$$

where p_f is the per-channel false detection probability, and the true-positive probability is:

$$P_{TP,maj} = \sum_{k=\lceil m/2 \rceil}^{m} \binom{m}{k} p_t^k (1-p_t)^{m-k},$$

with $p_t = 1 - p_f$ [3]. This framework minimizes assumptions, focusing on the ontological primitive of agent-triggered events.

4.2 Bypassing Traditional Constraints

MEBTI bypasses the constraints of Shannon, von Neumann, and Gabor by redefining information outside ensemble statistics or state uncertainty. Shannon's framework requires manipulating probability distributions, which the no-signaling theorem prohibits [3]. Von Neumann's entropy assumes quantum state mixedness, constrained by the Born rule [**?**]. Gabor's signal-based information is limited by physical degrees of freedom [10]. MEBTI's focus on single, agent-triggered events sidesteps these by not requiring statistical changes in marginals, instead emphasizing the detectability of a deliberate action. This redefinition challenges the no-signaling theorem's applicability to joint information encoded in single-run correlations.

4.3 Rebutting the Universality of the No-Signaling Theorem

On conceptual and logical grounds, MEBTI undermines the no-signaling theorem's claim to universality across all definitions of information. The theorem assumes information as a statistical construct, where signaling requires altering distant marginal probabilities [1]. MEBTI's single-event ontology shifts the question to whether an agent-triggered event can be detected remotely in a single run, a scenario not addressed by standard proofs. For a singlet state $|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, the joint probability $P(A, B|\theta_A, \theta_B) = \frac{1}{2}\sin^2(\theta_A - \theta_B)$ is fixed at entanglement, suggesting instantaneous correlation [2]. If Alice's trigger encodes a binary event in her measurement choice, Bob's detection of correlated outcomes could constitute information transfer under MEBTI, untested by current experiments.

4.4 Proposed Experiments

We propose experiments to test whether MEBTI's definition enables instantaneous information transmission via entanglement. Unlike Bell tests, which focus on ruling out FTL signaling by rapid detector switching [9], these experiments would prioritize detecting agent-triggered events. A possible protocol involves Alice encoding a binary event (e.g., choosing a measurement basis to represent $E = 0$ or 1) and Bob detecting the correlated outcome in a single run. By repeating trials and analyzing detection patterns, we could assess whether Bob reliably infers Alice's trigger, using MEBTI's majority-voting framework to ensure robustness [3]. Such experiments require novel theoretical models, potentially extending quantum information protocols [11], to explore whether entanglement supports instantaneous event detection without violating physical constraints.

5 Conclusion

The no-signaling theorem, while robust within standard quantum mechanics, assumes a statistical definition of information that MEBTI challenges [1**?**]. By redefining information as the reliable detection of agent-triggered binary events, MEBTI exposes the theorem's ontological limitations, rebutting its universal applicability on logical grounds. Within orthodox quantum mechanics, the theorem holds due to the uncontrollability of individual outcomes [2]. However, MEBTI suggests that new experiments, designed to test single-event detection via entanglement, could empirically validate or refute the possibility of instantaneous information transmission, potentially reshaping our understanding of quantum non-locality and information theory [3**?**].

6 Acknowledgments

The author expresses his gratitude to Grok [xAI] for assistance with research and the drafting of the article.

References

[1] Peres, A.: Quantum Theory: Concepts and Methods. Kluwer Academic Publishers, Dordrecht (1995). https://doi.org/10.1007/0-306-47120-5

[2] Bell, J.S.: On the einstein podolsky rosen paradox. Physics Physique Fizika **1**(3), 195–200 (1964) https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195

[3] Shannon, C.E.: A mathematical theory of communication. Bell System Technical Journal **27**(3), 379–423 (1948) https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

[4] Einstein, A., Podolsky, B., Rosen, N.: Can quantum-mechanical description of physical reality be considered complete? Physical Review **47**(10), 777–780 (1935) https://doi.org/10.1103/PhysRev.47.777

[5] Aspect, A., Dalibard, J., Roger, G.: Experimental test of bell's inequalities using time-varying analyzers. Physical Review Letters **49**(25), 1804–1807 (1982) https://doi.org/10.1103/PhysRevLett.49.1804

[6] Giustina, M., *et al.*: Significant-loophole-free test of bell's theorem with entangled photons. Physical Review Letters **115**(25), 250401 (2015) https://doi.org/10.1103/PhysRevLett.115.250401

[7] Hensen, B., *et al.*: Loophole-free bell inequality violation using electron spins separated by 1.3 kilometres. Nature **526**(7575), 682–686 (2015) https://doi.org/10.1038/nature15759

[8] Ghirardi, G.C., Rimini, A., Weber, T.: A model for a unified quantum description of microscopic and macroscopic systems. Il Nuovo Cimento B **59**(1), 263–277

(1980) https://doi.org/10.1007/BF02728053

[9] Weihs, G., *et al.*: Violation of bell's inequality under strict einstein locality conditions. Physical Review Letters **81**(22), 5039–5043 (1998) https://doi.org/10.1103/PhysRevLett.81.5039

[10] Gabor, D.: Theory of communication. Journal of the Institution of Electrical Engineers **93**(26), 429–457 (1946) https://doi.org/10.1049/ji-3-2.1946.0074

[11] Nielsen, M.A., Chuang, I.L.: Quantum Computation and Quantum Information. Cambridge University Press, Cambridge (2000). https://doi.org/10.1017/CBO9780511976667